

TwinLine

Strategic Growth Proposal

Accelerating Operational Excellence



Team InnovAstra

Executive Summary

Bridges the gap between siloed plant systems uniting into a single, real-time shared digital twin.

Transitions manufacturing operations from reactive firefighting to proactive prevention

Eliminates late detection and manual investigations by predicting bottlenecks and WIP buildup

Prevents quality escapes and costly line stoppages through early defect prediction

Delivers proven economic impact, including a 5-12% increase in line availability, 15-25% lower unplanned downtime, and typical savings of ₹15–30 Crore/year with a 6–12 month payback

Core Strategic Pillars

Unified Real-Time Visibility

- Breaks down historical data silos by integrating MES, SCADA, ERP, and QMS platforms into a single, cohesive digital twin.
- Continuously ingests live shopfloor signals via GCP Pub/Sub and Dataflow to maintain an up-to-date station topology model.

Proactive Congestion & Bottleneck Mitigation

- Predicts line congestion and WIP buildup before cycle times exceed takt targets, moving beyond late time-series forecasting.
- Gives production planners critical lead time to clear bottlenecks and prevent downstream delays, scrap, or line stoppages.

Quality Assurance & Defect Prevention

- Eliminates quality escapes usually caught only at final inspection by analyzing live machine and process parameters.
- Flags high-risk vehicles and anomalies early to prevent defects from cascading downstream and reducing rework rates.

Material Flow & Inventory Optimization

- Cross-references live component consumption against inventory levels and supplier schedules to anticipate material shortages.
- Optimizes stock levels to prevent production halts while minimizing excess inventory carrying costs.

Automated Diagnostics & Root-Cause Analysis (RCA)

- Replaces slow manual investigations with automated correlation engines that link process, machine, maintenance, quality, and material data.
- Generates ranked, explained contributing factors and actionable recommendations to help teams resolve operational risks swiftly.

Technical Novelty

Graph aware reasoning

Reactive + proactive intelligence

Multi-source fusion

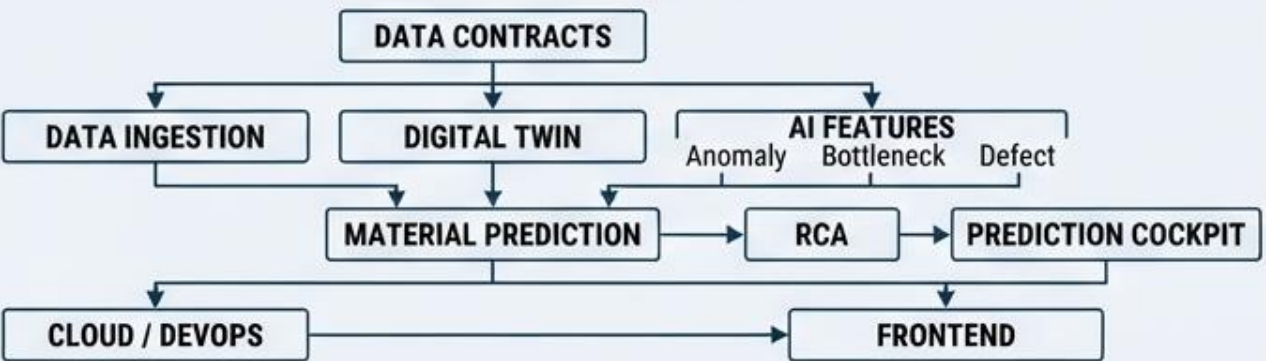
Explainable decision support

Human-in-the-loop design

TWINLINE 10-SPRINT PARALLEL DELIVERY ROADMAP

	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
CORE PLATFORM / DATA	Define event schema + data contracts	CSV/MES event ingestion	/ingest API + persistence	◆ Unified event pipeline	Data quality/feature pipeline	Quality + process data	Inventory + consumption data	Maintenance + material data	◆ Unified feature/state layer	◆ End-to-end integration
DIGITAL TWIN	Define station topology	Station master + NetworkX model	Live station state	◆ Live Digital Twin MVP	State enrichment	Vehicle/station correlation	Material-state linkage	Cross-station relationships	◆ Cross-signal correlation	◆ TwinLine v1
AI / ML	Define ML features & targets	Generate training/synthetic datasets	Anomaly model	Bottleneck model	◆ Anomaly Detection MVP	◆ Bottleneck Prediction MVP	◆ Defect Prediction	◆ Material/Stockout Prediction	◆ Root Cause Analysis	◆ All models integrated
UI / EXPERIENCE	Dashboard wireframe	Streamlit skeleton	Live station visualization	Digital Twin dashboard	Alerts + KPI cards	Prediction visualization	Risk dashboard	Intervention views	Prediction Cockpit	◆ Final Cockpit
DEVOPS / CLOUD	GCP architecture + repo	Cloud Run + Cloud SQL + CI/CD	Logging + Secret Manager	Performance testing	Model packaging	Model deployment	Monitoring	Reliability testing	Security + observability	◆ Demo / scale testing

STREAMLINED DEPENDENCY MAP



KEY EXECUTION PHASES

	S1-S2: FOUNDATION	Parallel setup & data contracts. Schemas, mock datasets, GCP repo.
	S3-S4: MVP ACTIVATION	First vertical slice & Live Twin. FastAPI ingestion, live station state.
	S5-S8: CAPABILITY EXPANSION	Parallel AI models & UI cards. Stockout models, risk dashboards.
	S9-S10: INTEGRATION & POLISH	Cross-signal synthesis & deployment. Unified RCA, final cockpit.

POC Scope Alignment

MVP Element	TwinLine Proposal Alignment
Data sources	Shopfloor systems + MES-style events + reference/master data
Digital Twin	Live graph of stations, vehicles, WIP and process dependencies
AI	Anomaly detection + proactive bottleneck prediction
Users	Plant Operator, Maintenance Engineer, Production Planner
Dashboards	Live Line Dashboard + Prediction Cockpit
Business outcomes	Downtime ↓ Throughput ↑ WIP ↓

Key risks and Mitigation

